

Fabrice Harel-Canada

RESEARCHER · SOFTWARE ENGINEER

☎ (+1) 954-805-5735 | ✉ fabricehc@cs.ucla.edu | 🌐 fabrice.harel-canada.com | 📱 fabriceyh

Biography

Recent UCLA Computer Science PhD specializing in pioneering robust and holistic evaluation frameworks for AI systems. My doctoral research includes award-winning work on human-centric LLM quality assessment, novel methods in AI testing, data augmentation (e.g., silybvariant transformations), content provenance (watermarking), and most recently NLP applied to substance use epidemiology.

Professional Experience

Kairno

FOUNDER AND CEO

May 2025 - Jan 2026

- Developed and deployed a full-stack system (@ kairno.com) that automates rigorous academic peer review, ensuring compliance with field-specific reporting guidelines and best practices.

Verox Tech

Supervisor: Michael Simhai

PROJECT MANAGER + DEVELOPER TEAM LEAD

July 2015 – Sept 2018

- Spearheaded the end-to-end development and launch of a cloud-native insurance platform, managing a cross-functional team of 23 engineers and designers.
- Architected scalable C# Web APIs and optimized complex T-SQL procedures, maintaining high performance for high-concurrency database transactions.
- Led a mission-critical migration of legacy data across three disparate systems, ensuring 100% data integrity and zero downtime during the transition.
- Responsible for producing, coordinating and reviewing nearly every aspect of the SDLC: code reviews and code optimization, requirements and spec generation, database design, technical documentation, UI design standards, and deployment scripts.

People's Trust Insurance

Supervisor: Scott Thompson

BUSINESS + DATA ANALYST

Dec 2012 – Sept 2018

- Served as lead data strategist for the executive suite, delivering 100+ high-impact analytical reports that drove multi-million dollar business decisions.
- Developed numerous front-end / back-end application features that were successfully deployed into production to automate tedious and time-intensive workflows.
- Bridged the gap between business stakeholders and engineering teams across 60+ agile sprints, overseeing the transition to a modernized policy and claim management ecosystem.

Cooperative Living Organization

PRESIDENT

Dec 2010 – May 2012

- Directed organizational operations by managing 17 department heads and implementing standardized policy frameworks.
- Optimized revenue streams and marketing, increasing house occupancy from 75% to 100% capacity and driving a 57% increase in net income.

Research Experience

The Wellness Center @ LA General Hospital

Supervisor: Rebecca Trotsky

APPLIED RESEARCH ENGINEER

July 2025 - Present

- Engineer and deploy fine-tuned LLMs to extract substance use history from clinical notes, achieving high-precision classification between active and historical usage.
- Architected a large-scale data pipeline to perform epidemiological statistical analysis on 1M+ electronic health records to identify longitudinal trends in substance use.

University of California, Los Angeles

Supervisor: Chelsea Shover

GRADUATE STUDENT RESEARCHER

Dec 2023 - July 2025

- Developed a context-aware drug misuse detection system ($F1 > 0.95$), significantly reducing false positives in clinical screening across multiple drug classes.
- Helped engineer a predictive geospatial model to forecast overdose hotspots in Los Angeles County, integrating multi-modal data to inform public health intervention strategies.

University of California, Los Angeles

GRADUATE STUDENT RESEARCHER

Supervisor: *Miryung Kim*

June 2019 - Dec 2023

- Validated the real-world robustness of AI watermarking through large-scale experimentation, overturning theoretical assumptions about watermark erasability and demonstrating a 90% retention rate against state-of-the-art attacks.
- Developed a theoretically grounded framework to assess the psychological depth of LLMs, revealing that modern LLMs match or exceed the abilities of advanced human short story writers.
- Created automated evaluation metrics for natural language generation systems, including measures of linguistic diversity – e.g. semantic, syntactic, morphological, phonological – and as well as general engagement.
- Designed and open-sourced a new class of data augmentations called sibylvariant transforms for enhancing NLP model robustness.
- Assessed the reliability of structural testing metrics, such as neuron coverage, in evaluating deep learning models.
- Developed diversity-enhancing extensions for adversarial attack algorithms to improve robustness testing.

University of California, Los Angeles

RESEARCH ASSISTANT

Supervisor: *Judson Caskey*

July 2020 - Oct 2020

- Leveraged BERT-based transformers and LDA to execute sentiment analysis and topic modeling on a massive dataset of 1M+ Amazon and Glassdoor reviews.

Education

University of California, Los Angeles (UCLA)

PHD | COMPUTER SCIENCE

Advisor: *Miryung Kim*

Jan 2020 - June 2025

- 3.92 / 4.00 GPA
- Majors: Artificial Intelligence + Software Engineering | Minor: Data Science Computing
- Thesis: *From Metrics to Meaning: Advancing Evaluation Frameworks for Robust and Human-Centric AI*

University of California, Los Angeles (UCLA)

MS | COMPUTER SCIENCE

Advisor: *Miryung Kim*

Sept 2018 - Dec 2019

- Major: Artificial Intelligence
- Thesis: *Is Neuron Coverage a Meaningful Measure for Testing Deep Neural Networks?*

University of Florida (UF)

BS | BUSINESS ADMINISTRATION

Aug 2008 - May 2012

- Majors: Management + Information Systems and Operations Management
- Minors: Entrepreneurship + Finance
- Thesis: *The Viability of the Student-Run Housing Cooperative: A Case Study of the Cooperative Living Organization*

Skills

Programming	python, sql, java, c#, linq, html, css, js, scala, prolog, prolog, $\LaTeX$
Frameworks	huggingface, pytorch, tensorflow, keras, spark, numpy, pandas, scipy, sklearn, opencv, matplotlib, seaborn
Tools	ssms, ssis, ssrs, tf, azure devops, visual studio, excel, aws, gcp, exchange server, moqups